FORMATIVE ASSESSMENT – 3 (FA-3)

MATHEMATICS PROJECT WORK

Topic: Criteria for Congruence of Triangles (SAS & SSS)

PRELIMINARY INFORMATION

Name of the Student: .....

Class & Section: **Class 9 ( )**

Roll Number: .....

Name of the Lesson: **Triangles**

Name of the Guide Teacher: .....

AIM

To construct pairs of triangles with specified corresponding sides and included angles, and verify their congruence by superposition and direct measurement.

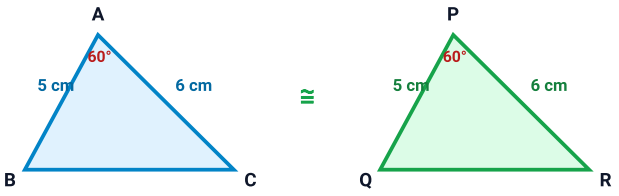
MATERIAL REQUIRED

- Drawing Paper / Chart
- Geometry Ruler Scale
- Protractor & Compass
- Pencil & Eraser
- Scissors

PROCEDURE

1. **Construction:** Construct triangle ABC with AB = 5 cm, AC = 6 cm, and included angle ∠A = 60°.
2. **Replication:** Construct a second triangle PQR with PQ = 5 cm, PR = 6 cm, and included angle ∠P = 60°.
3. **Superposition & Measurement:** Cut out triangle PQR and place it over triangle ABC to test exact overlap and verify matching parts.

GEOMETRIC FIGURE (CONGRUENT TRIANGLES ABC AND PQR)



OBSERVATION AND MEASUREMENT TABLE

S.No.	Corresponding Parts	ΔABC Value	ΔPQR Value	Observation
1	Side 1 (Base/Arm)	AB = 5.0 cm	PQ = 5.0 cm	**Equal (AB = PQ)**
2	Included Angle	∠A = 60°	∠P = 60°	**Equal (∠A = ∠P)**
3	Side 2 (Arm)	AC = 6.0 cm	PR = 6.0 cm	**Equal (AC = PR)**
4	Remaining Side (BC, QR)	BC = 5.6 cm	QR = 5.6 cm	**Equal (By CPCTC)**

CONCLUSION

Triangle PQR completely covers triangle ABC when superimposed. Hence, when two sides and the included angle of one triangle are equal to those of another, the triangles are congruent by the SAS criterion (ΔABC ≅ ΔPQR).

Signature of the Student

Date: _____

Signature of the Guide Teacher

Date: _____